

7. Tom competes in a 400 m training race. His split times to travel each 100 m are measured. His pulse rate is also measured in beats per minute (bpm) before the race starts and at every 100 m interval. The measurements are shown in **Table 1** below.

Table 1

Distance (m)	0	100	200	300	400
Time (s)	0	13	27	43	58
Pulse rate (bpm)	78	133	132	134	133

Use the data in the table and your knowledge to answer the following questions.

- (a) (i) State Tom's resting pulse rate. bpm [1]
- (ii) One minute after the race ended, Tom's pulse rate had dropped to 122.

Calculate **how much longer** it will be for his pulse to return to his resting rate, assuming it continues to drop at the same rate. [3]

time = minutes

- (iii) Use the equation

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

and the information in **Table 1** to calculate Tom's mean speed over the race. [2]

mean speed = m/s



- (iv) Use the information in **Table 1** to complete **Table 2** to show the order of running each 100 m section from fastest to slowest. The fastest 100 m section has already been filled in. [2]

Table 2

Fastest section



Slowest section

0 to 100 m
..... to m
..... to m
..... to m

- (b) Tom then trains for 6 months. **Complete Table 3** below by estimating Tom's pulse rate every 100 m in a 400 m race after training. [2]

Table 3

Distance (m)	0	100	200	300	400
Pulse rate (bpm)					

